Enamel and Investing layers of Enamel

PP COMPILATION

Q. The periodic changes in prism direction give rise to a banding pattern termed as: (Past paper 2017,19,23 forward)
a) Non prismatic enamel
b) Hunter-Schreger band
c) Cross-Striations
d) Enamel Striae.
e) Incremental lines

Q. Regarding the ionic exchange between enamel surface of enamel and oral fluid environment. The following is the best statement: (Past paper 2020 forward)
A. Does not take place once enamel surface and environment?
B. Stops after 2 years of eruption
C. Continues till adult life
D. Continues throughout life
E. only restricted to Acidic pH.

Q. Non-prismatic enamel differs from prismatic enamel in its structure and composition. Are the enamel crystallites oriented in non-prismatic enamel? (Past paper 2016,17,20 forward)
A) Parallel to the enamel-dentin junction
B) Randomly oriented
C) Parallel to each other and at right angles to the surface
D) Spirally arranged
E) Tangential to the surface

Q. Which pattern of enamel prisms predominates in human enamel in cross-section and is characterized by a keyhole shape with distinct head and tail regions? (Past paper 2020 forward)
A) Pattern 1 B) Pattern 2 C) Pattern 3 D) Pattern 4 E) Pattern 5

Q. What is the degree of deviation of crystallites at the prism boundaries compared to the core that results in increased microporosity? (Past paper 2020 forward)
A) 10–20° B) 20–30° C) 40–60° D) 70–90° E) 90–120°

Q. Which of the following is a characteristic of surface enamel compared to subsurface enamel? (Past paper 2020 forward)
A) More porous and soluble
B) Contains more carbonate
C) Harder and less radio-opaque
D) Richer in fluoride and less porous
E) Lower mineral content

Q. Following areas in enamel are potential spots of weakness in the hard tissue: (Past paper 2020 forward)
A) Enamel broaches B) Focal holes C) Enamel caps D) Enamel cracks E) Perikymata grooves

Q. The band of enamel prisms that are cut longitudinally are known as: (Past paper 2020 forward)
A) Parazones B) Gnarled enamel C) Enamel tufts D) Enamel lamellae E) Diazones

Q. What is the organic layer of salivary origin that forms on the enamel surface called? (Past paper 2020 forward)
a. Reduced enamel epithelium b. Primary enamel cuticle c. Acquired pellicle d. Junctional epithelium e. Enamel matrix

Q. Primary teeth appear whiter than permanent teeth because: (Past paper 2020 forward)

Q. Incorrect statement regarding enamel tufts is : (Past paper 2020 forward)
a. present through out the length of enamel
b. present at inner third of enamel
c. best seen at transverse section
d. contain protein tuftlin
e. it is hypomineralized

Q. Crown of the tooth is covered by protective layers throughout its life; Nasmyth's membrane is the organic layer that covers the crown immediately after its emergence in the oral cavity is the formed of: (Past paper 2020 forward)
a. Primary enamel cuticle and reduced enamel epithelium
b. Salivary proteins
c. Basal Lamina of outer enamel epithelium
d. Combination of salivary proteins and microorganisms
e. Secreted by ameloblast

Q. Aprismatic or prismless enamel are present at: (Past paper 2020 forward)
a. whole enamel thickness
b. Dentinoenamel junction and outer most enamel
c. only present in deciduous dentition
d. outer surface of enamel
e. Dentinoenamel junction

Q. Metaloproteinases are used in which stage of Amelogenesis: (Past paper 2020 forward)
a) Histological stage b) Differentiation stage c) Maturation d) Formation e) Differentiation & proliferation

Q. Young enamel protein has: (Past paper 2020 forward)
a) 90% amelogenins & 10% non-amelogenins b) 90% non amelogenin & 10 % amelogenin c) Only 90 % non amelogenin d) Only 90% amelogenin e) 50% amelogenin & 50 % non-amelogenin.

Q. When enamel surface is viewed under the microscope, shallow furrows called perikymata are seen. These are likely due to: (Past paper 2019 forward)
a) Abrasion from the tooth brush
b) Attrition from occlusal forces
c) Crakes (Lamella) in enamel
d) Incremental formation in enamel
e) Organic deposits from saliva

Q. The complex arrangement of rods over cuspal area is known as: (Past paper 2016 forward)
a) Prism less enamel b) Pre natal enamel c) Post natal enamel d) Gnarled enamel e) Enamel lamellae

Q. In enamel the molecular arrangement within each unit cell of crystallite consists of: (Mid 2022)
a. OH group surrounded by three uniformly spaced phosphate ions further surrounded by Ca ions.
b. OH group surrounded by three uniformly spaced Ca ions and in turn surrounded by PO4 ions.
c. Water crystal surrounded by Ca and PO4 ions with equal spaces
d. Ca ion surrounded by OH ions
e. PO4 ions surrounded by equally placed Ca ions

Q. The substance which has been identified in cross striations and Hunter Schreger bands, although in little amount but may represent the remnants of cell membrane from development is: (Mid 2022)
a. Protein b. CHO c. Lipids d. Amino Acids e. Polysaccharides

Q. The Nano hardness and elastic modulus of prism boundaries were approximately: (Mid 2022)
a. 50 & 75% lower than those in within cores
b. 75 & 50 % greater than those in within cores
c. 75 & 50% lower than those in within cores
d. 50 & 75 % greater than those in within cores
e. Both are same without any difference

Q. Regarding, cemento enamel junction the following is the most abundant type found in human dentition: (Past paper 2020 forward)
a. cementum overlaps the cervical end of enamel. b. enamel and cementum do not meet c. enamel overlaps cementum. d. Cementum is absent completely at the cervical e. cementum meets the cervical end of enamel.

Q. The crown of the tooth is protected from external stimuli at every stage of its development. The soft tissue layer covering the unerupted crown of tooth is: (Past paper 2020 forward)
a) Reduced enamel epithelium b) Primary enamel cuticle c) Pellicle d) Plaque e) Calculus

Q. Smaller elevation on some portion of the crown produced by the extra formation of enamel is known as: (Mid 2022)
a. Cusps b. Cingulum c. Tubercle d. Ridge e. Lobe

Q. In chemical composition of dentine / enamel: Enamel = 96% inorganic. (Past paper 2020 forward)

Q. Nasmyth membrane is = reduced enamel epithelium + basal lamina. (Past paper 2020 forward)

Q. Basic unit of enamel responsible for its physical properties and geographical topology is = enamel prism. (Past paper 2020 forward)

Q. What enamel is more permeable between primary and permanent = primary enamel is more permeable. (Past paper 2020 forward)

Q. Prismless enamel is = parallel to each other and perpendicular to surface. (Past paper 2020 forward)

Q. Function of Tomes' process = Secretes enamel matrix. (Past paper 2020 forward)

Q. Following space is the potential space of enamel weakness = Enamel crack. (Past paper 2020 forward)

Q. Following is correct about prism = prism II and prism III boundaries are incomplete. (Past paper 2020 forward)

Q. Enamel prism cut longitudinally = parazone. (Past paper 2020 forward)

Q. Characteristics of enamel that is specie specific is? (Past paper 2020 forward)

Q. Which of the following is responsible for the formation of enamel prisms? (Past paper 2020 forward) a) ameloblast b) tomes process

Q. Temporary structures in enamel? (Past paper 2020 forward)

Q. Interprismatic rods are formed as a result of: (Past paper 2020 forward)

Q. Layer covering the newly erupted tooth which is: (Past paper 2020 forward)

Q. Highly mineralized and hardest tissue of body? ENAMEL. (Mid 2022)

Q. Enamel thickness = 2.5 mm. (Past paper

Q. Arrangement of prisms on occlusion = 20 or 60 degree. (Past paper

Q. Cross striations are best seen: (Past paper 2020 forward)

Q. The lipid content of the enamel is: (Past paper 2020 forward)

Q. The periodic changes in prism direction give rise to a banding pattern termed as: (Past paper 2020 forward)
Answer: b) Hunter-Schreger band

(Note: Optical phenomenon caused by alternating curvature/direction of prism groups; cross-striations represent daily incremental growth lines).

Q. Regarding the ionic exchange between enamel surface of enamel and oral fluid environment. The following is the best statement: (Past paper 2020 forward)
Answer: D. Continues throughout life

(Note: Mature enamel functions as a semipermeable membrane allowing dynamic ionic exchange—such as fluoride uptake, demineralization, and remineralization—with saliva across the lifespan).

Q. Non-prismatic enamel differs from prismatic enamel in its structure and composition. Are the enamel crystallites oriented in non-prismatic enamel? (Past paper 2020 forward)
Answer: C) Parallel to each other and at right angles to the surface

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Q. Which pattern of enamel prisms predominates in human enamel in cross-section and is characterized by a keyhole shape with distinct head and tail regions? (Past paper 2020 forward)
Answer: C) Pattern 3

Q. What is the degree of deviation of crystallites at the prism boundaries compared to the core that results in increased microporosity? (Past paper 2020 forward)
Answer: C) 40–60°

Q. Which of the following is a characteristic of surface enamel compared to subsurface enamel? (Past paper 2020 forward)
Answer: D) Richer in fluoride and less porous

Q. Following areas in enamel are potential spots of weakness in the hard tissue: (Past paper 2020 forward)
Answer: D) Enamel cracks

Q. The band of enamel prisms that are cut longitudinally are known as: (Past paper 2020 forward)
Answer: A) Parazones

(Note: Longitudinally sectioned prisms appear light as parazones; transversely cut prisms appear dark as diazones).

Q. What is the organic layer of salivary origin that forms on the enamel surface called? (Past paper 2020 forward)
Answer: c. Acquired pellicle

Q. Primary teeth appear whiter than permanent teeth because: (Past paper 2020 forward)
Answer: a. They have a thinner and more opaque/consistent enamel layer that reflects more light, combined with less underlying bulk dentin.

Q. Incorrect statement regarding enamel tufts is : (Past paper 2020 forward)
Answer: a. present through out the length of enamel

(Note: Enamel tufts extend only through the inner 1/3 of the enamel thickness from the DEJ; enamel lamellae extend through the full thickness).

Q. Crown of the tooth is covered by protective layers throughout its life; Nasmyth's membrane is the organic layer that covers the crown immediately

after its emergence in the oral cavity is the formed of: (Past paper 2020 forward)
Answer: a. Primary enamel cuticle and reduced enamel epithelium

Q. Aprismatic or prismless enamel are present at: (Past paper 2020 forward)
Answer: b. Dentinoenamel junction and outer most enamel

Q. Metaloproteinases are used in which stage of Amelogenesis: (Past paper 2020 forward)
Answer: c) Maturation

(Note: MMP-20/kallikrein-4 degrade organic enamel matrix during the maturation stage to allow crystal growth).

Q. Young enamel protein has: (Past paper 2020 forward)
Answer: a) 90% amelogenins & 10% non-amelogenins

Q. When enamel surface is viewed under the microscope, shallow furrows called perikymata are seen. These are likely due to: (Past paper 2020 forward)
Answer: d) Incremental formation in enamel

(Note: Perikymata are the external surface manifestations/grooves of the Striae of Retzius).

Q. The complex arrangement of rods over cuspal area is known as: (Past paper 2020 forward)
Answer: d) Gnarled enamel

Q. In enamel the molecular arrangement within each unit cell of crystallite consists of: (Mid 2022)
Answer: b. OH group surrounded by three uniformly spaced Ca ions and in turn surrounded by PO4 ions.

(Note: In the hydroxyapatite hexagonal lattice, central hydroxyl ions are coordinated by concentric rings of calcium and phosphate ions).

Q. The substance which has been identified in cross striations and Hunter Schreger bands, although in little amount but may represent the remnants of cell membrane from development is: (Mid 2022)
Answer: c. Lipids

Q. The Nano hardness and elastic modulus of prism boundaries were approximately: (Mid 2022)
Answer: c. 75 & 50% lower than those in within cores
(Note: Due to abrupt crystallite angulation and higher organic/water content at the prism boundaries).

Q. Regarding, cemento enamel junction the following is the most abundant type found in human dentition: (Past paper 2020 forward)
Answer: a. cementum overlaps the cervical end of enamel

(Note: Occurs in approximately 60% of cases).

Q. The crown of the tooth is protected from external stimuli at every stage of its development. The soft tissue layer covering the unerupted crown of tooth is: (Past paper 2020 forward)
Answer: a) Reduced enamel epithelium
(Note: Formed by the post-secretory ameloblasts and the collapsed outer layers of the enamel organ).

Q. Smaller elevation on some portion of the crown produced by the extra formation of enamel is known as: (Mid 2022)
Answer: c. Tubercle

Q. In chemical composition of dentine / enamel: Enamel = 96% inorganic. (Past paper 2020 forward)
Answer: 96% inorganic by weight (86% by volume), 1% organic, and 3% water.

Q. Nasmyth membrane is = reduced enamel epithelium + basal lamina. (Past paper 2020 forward)
Answer: Reduced enamel epithelium (REE) + primary enamel cuticle (basal lamina secreted by ameloblasts).

Q. Basic unit of enamel responsible for its physical properties and geographical topology is = enamel prism. (Past paper 2020 forward)
Answer: Enamel prism (enamel rod).

Q. What enamel is more permeable between primary and permanent = primary enamel is more permeable. (Past paper 2020 forward)
Answer: Primary enamel (due to lower overall mineral density and higher pore volume).

Q. Prismless enamel is = parallel to each other and perpendicular to surface. (Past paper 2020 forward)
Answer: Formed without Tomes' processes; crystallites are aligned parallel to one another and perpendicular to the tooth surface.

Q. Function of Tomes' process = Secretes enamel matrix. (Past paper 2020 forward)
Answer: Directs the orientation of crystallites and secretes the enamel matrix in rod and interrod configurations.

Q. Following space is the potential space of enamel weakness = Enamel crack. (Past paper 2020 forward)

Answer: Enamel crack (lamellae/faults).

Q. Following is correct about prism = prism II and prism III boundaries are incomplete. (Past paper 2020 forward)
Answer: Pattern III (keyhole pattern) exhibits incomplete prism sheaths at the cervical/tail aspect.

Q. Enamel prism cut longitudinally = parazone. (Past paper 2020 forward)

Answer: Parazone (light bands in Hunter-Schreger phenomenon).

Q. Characteristics of enamel that is specie specific is? (Past paper 2020 forward)
Answer: Prism cross-sectional packing pattern (Patterns 1, 2, or 3) and Hunter-Schreger band decussation configuration.

Q. Which of the following is responsible for the formation of enamel prisms? (Past paper 2020 forward)
Answer: b) Tomes' process

Q. Temporary structures in enamel? (Past paper 2020 forward)
Answer: Enamel knot and Enamel cord (temporary signaling centers during the cap/bell stages).

Q. Interprismatic rods are formed as a result of: (Past paper 2020 forward)
Answer: Secretion by the proximal portion / interprocess portion of Tomes' processes at an angle to the main rod axis.

Q. Layer covering the newly erupted tooth which is: (Past paper 2020 forward)
Answer: Nasmyth's membrane (Primary enamel cuticle + REE remnant).

Q. Highly mineralized and hardest tissue of body? ENAMEL. (Mid 2022)
Answer: Enamel.

Q. Enamel thickness = 2.5 mm. (Past paper 2020 forward)
Answer: ~2.0 to 2.5 mm (maximum over the cusps and incisal edges of permanent teeth).

Q. Arrangement of prisms on occlusion = 20 or 60 degree. (Past paper 2020 forward)

Answer: Prisms run generally perpendicular to the DEJ and occlusal/incisal surface (crystallite deviation at rod boundaries varies 40°–60°).

Q. Cross striations are best seen: (Past paper 2020 forward)
Answer: In longitudinal ground sections under polarized light or phase-contrast microscopy.

Q. The lipid content of the enamel is: (Past paper 2020 forward)
Answer: Approximately 0.6% of the total enamel weight (~40% of the total organic fraction of mature enamel).